

# Sky-Guard Radar Detection System

360° low-altitude UAV detection and precision EO cueing



## Overview

A Ku-band radar subsystem composed of the radar unit, motorised turntable, and operator terminal software. It detects and tracks low-altitude UAV targets, outputs real-time 3D target parameters, and provides precise cueing for electro-optical, jamming, and spoofing subsystems.

## Key Capabilities

- 360° mechanical azimuth scan with electronic elevation coverage from 0° to 30°.
- Outputs range, azimuth, altitude, speed, longitude, and latitude in real time.
- Supports tripod deployment, vehicle-roof mobile monitoring, or fixed rooftop stations.
- Fully coherent pulse-Doppler processing, monopulse angle measurement, MTD integration, and CFAR detection.

## Technical Specifications

|                          |                                            |
|--------------------------|--------------------------------------------|
| Frequency Band           | Ku, 16 GHz ± 100 MHz                       |
| Azimuth / Elevation      | 360° mechanical / ≥ 30° electronic         |
| Max Detection Range      | ≥ 3 km (DJI Phantom 3/4, Pd = 85%)         |
| Minimum Detection Range  | ≤ 250 m                                    |
| Minimum Detectable Speed | ≤ 1.5 m/s                                  |
| 3D Accuracy              | Range ≤ 10 m; azimuth/elevation ≤ 0.5° RMS |
| Refresh Rate             | 3 s per 360° scan                          |
| Weight / Power           | ≤ 30 kg; ≤ 150 W including turntable       |

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